

Name:

BUID:

Workshop 4: Subspaces, Dimension, Ranks-Nullity

1. Let A denote the matrix $\begin{bmatrix} 1 & -2 & 19 & 0 & -4 \\ 1 & 0 & 9 & 1 & 1 \\ 1 & -1 & 14 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$

- (a) Determine a basis for $\text{Col } A$. (b) Determine a basis for $\text{Nul } A$. (c) Determine the dimension of $\text{Col } A$.
(d) Determine the dimension of $\text{Nul } A$.

2. Let A be the matrix $\begin{bmatrix} 1 & -4 & 3 & -3 \\ -2 & 8 & -6 & 7 \end{bmatrix}$.

- (a) Determine a basis for $\text{Col } A$. (b) Determine a basis for $\text{Nul } A$. (c) Determine the dimension of $\text{Col } A$.
(d) Determine the dimension of $\text{Nul } A$.

3. Determine the coordinate vector $\mathbf{v} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ relative to the basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ of $\mathbb{R}^2$ (i.e. determine $[\mathbf{v}]_{\mathcal{B}}$).

4. Determine the coordinate vector $\mathbf{v} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$ relative to the basis $\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$ of a subspace of $\mathbb{R}^3$ (i.e., determine $[\mathbf{v}]_{\mathcal{B}}$).

5. Suppose that $[\mathbf{u}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$ where $\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

6. Let $A \in \mathbb{R}^{8 \times 11}$ be a matrix such that $\dim \text{Nul } A = 5$. Determine $\text{rank } A$.

7. Let $A \in \mathbb{R}^{9 \times 15}$ such that A^T has linearly independent columns. Determine $\dim \text{Nul } A$.

8. Show that if U and V are subspaces of $\mathbb{R}^n$, then so is $U \cap V = \{\mathbf{v} : \mathbf{v} \in U \text{ and } \mathbf{v} \in V\}$.

Write your solutions below and on the back on this page.

1. (a) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

(b) $x_1 = -9x_3$
 $x_2 = 5x_3 - 2x_5$
 x_3 is free
 $x_4 = -x_5$
 x_5 is free

$\begin{bmatrix} -9 \\ 5 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}$

(c) 3

(d) 2

2. (a) $\begin{bmatrix} 1 & -4 & 3 & -3 \\ -2 & 8 & -6 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 3 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 7 \end{bmatrix}$

(b) $x_1 = 4x_2 - 3x_3$
 x_2 is free
 x_3 is free
 $x_4 = 0$
 $\begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

(c) 2 (d) 2

3. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \end{bmatrix}$

$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$

$-2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix} \checkmark$

4. $\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 3 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -3 \\ 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}_{\mathcal{B}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$2 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$

Continue your work here.

$$5. [\vec{u}]_{\beta} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \Rightarrow \vec{u} = 2 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 6-2 \\ 12+0 \\ 4+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ 6 \end{bmatrix}$$

$$6. \text{rank}(A) + \underset{\substack{= \\ 5}}{\dim(\text{Nul}(A))} = 11 \Rightarrow \text{rank}(A) = 6$$

$$7. A^T \text{ has LI columns} \Rightarrow A \text{ has 9 pivots} \Rightarrow \text{rank}(A) = 9$$

$$\underset{\substack{= \\ 9}}{\text{rank}(A)} + \dim(\text{Nul}(A)) = 15$$

$$\dim(\text{Nul}(A)) = 6$$

8. Closed under addition:

$$\begin{array}{l} \vec{u} \in U \cap V \Rightarrow \vec{u} \in U, \vec{r} \in U \Rightarrow \vec{u} + \vec{r} \in U \\ \vec{r} \in U \cap V \Rightarrow \vec{u} \in V, \vec{r} \in U \Rightarrow \vec{u} + \vec{r} \in V \end{array} \Rightarrow \vec{u} + \vec{r} \in U \cap V$$

Closed under scalar:

$$\begin{array}{l} \vec{u} \in U \cap V \Rightarrow \vec{u} \in U \Rightarrow c\vec{u} \in U \\ \vec{u} \in V \Rightarrow c\vec{u} \in V \end{array} \Rightarrow c\vec{u} \in U \cap V$$